

Question Bank
Part I
Complex Variables
and Linear Algebra
(BMAT201L)

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Vellore Institute of Technology
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BMAT201L	Complex Variables and Linear Algebra	L	T	P	C
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Pre-requisite	BMAT102L	Syllabus version			
		1.0			
Course Objectives					
1. To present comprehensive, compact, and integrated treatment of one of the most important branches of applied mathematics namely Complex variables to the engineers and the scientists. 2. To present comprehensive, compact, and integrated treatment of another most important branches of applied mathematics namely Linear Algebra to the engineers and the scientists. 3. To provide students with a framework of the concepts that will help them to analyse deeply about many complex problems.					
Course Outcomes					
At the end of the course the student should be able to					
1. Construct analytic functions and find complex potential of fluid flow and electric fields. 2. Find the image of straight lines by elementary transformations and to express analytic functions in power series. 3. Evaluate real integrals using techniques of contour integration. 4. Use the power of inner product and norm for analysis. 5. Use matrices and transformations for solving engineering problems.					
Module:1	Analytic Functions	7hours			
Complex variable - Analytic functions and Cauchy – Riemann equations; Laplace equation and Harmonic functions; Construction of Harmonic conjugate and analytic functions; Applications of analytic functions to fluid-flow and electric field problems.					
Module:2	Conformal and Bilinear transformations	7 hours			
Conformal mapping - Elementary transformations; Translation, Magnification, Rotation, Inversion; Exponential and Square transformations ($w = e^z$, z^2); Bilinear transformation; Cross-ratio-Images of the regions bounded by straight lines under the above transformations;					
Module:3	Complex Integration	7 hours			
Functions given by Power Series - Taylor and Laurent series-Singularities - Poles – Residues; Integration of a complex function along a contour; Statements of Cauchy-Goursat theorem- Cauchy's integral formula-Cauchy's residue theorem-Evaluation of real integrals- Indented contour integral.					
Module:4	Vector Spaces	6 hours			
Vector space – subspace; linear combination - span - linearly dependent – Independent – bases; Dimensions; Finite dimensional vector space. Row and column spaces; Rank and nullity.					
Module:5	Linear Transformations	6 hours			
Linear transformations – Basic properties; Invertible linear transformation; Matrices of linear transformations; Vector space of linear transformations; Change of bases; Similarity.					
Module:6	Inner Product Spaces	5 hours			
Dot products and inner products; Lengths and angles of vectors; Matrix representations of inner products; Gram - Schmidt – Orthogonalization.					
Module:7	Matrices and System of Equations	5 hours			
Eigenvalues and Eigen vectors; Properties of Eigenvalues and Eigen vectors; Cayley-Hamilton theorem; System of linear equations; Gaussian elimination and Gauss Jordan methods.					
Module:8	Contemporary issues:	2 hours			

About the Question Bank

The Question Bank (Part 1) consists of three modules. The first module is about the analytic functions and their applications. The second module is about conformal mappings, bilinear transformations and their graphical interpretation. The third module deals with contour integration and computing real integrals through the calculus of residues.

In every module, some preliminaries are provided section-wise as a quick reference, which help the learners solve the exercise problems smoothly. The exercises completely cover the syllabus contents. Necessary hints are also given along with the answers. At the end of each module, a glossary of terms is also provided. Some useful references are also appended at the end.

The Question Bank is compiled with at most care, though it is a monumental task so that the aspiring learners get complete understanding of the course content. Valuable suggestions and constructive criticism in this regard is highly appreciable.

- The Author

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Module 1

Analytic Functions

1.1 Analytic Functions

Definition 1.1 (Analytic Function): Let $\mathcal{D} \subset \mathbb{C}$ and $z_0 \in \mathcal{D}$. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ is said to be *analytic* at $z_0 \in \mathcal{D}$, if it is differentiable on some neighborhood of z_0 . We say that f is *analytic* on $\mathcal{D}$, if it is analytic at every point of $\mathcal{D}$.

Nowhere differentiable functions are nowhere analytic.

Theorem 1.1 (Examining the Analyticity of Complex Functions): Consider $f(z) = u(z) + iv(z)$ and $z_0 = (x_0, y_0) \in \mathbb{C}$. Suppose that

(a) the partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous, and

(b) satisfy

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (1.1)$$

at every point of some neighborhood of (x_0, y_0) . Then f is analytic at z_0 . The derivative of f is

$$f'(z_0) = \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}. \quad (1.2)$$

If the Cauchy-Riemann equations (1.1) are not satisfied at a point, f cannot be differentiable and hence is not analytic at that point. $\square$

Exercise 1.1: Explain why the following functions $f(z)$ are nowhere analytic:

(a) x

(b) y

(c) $x - iy$

(d) $x(3 + 2iy^2)$

(e) $-x + iy + 5$

(f) $x^3 - xy^2 - 2x^2y^2 + i(2x^2y - y^3)$

(g) $x^3 + 3xy^2 - x + i(y^3 + 3x^2 - y)$.

Answers:

In (a), (b), (c), (d) and (e): The Cauchy-Riemann equations do not hold good at all points z of the complex plane.

In (f) and (g): The Cauchy-Riemann equations are satisfied only at the origin and $f'(0) = 0$.

However, in (a)-(g), $\exists$ no neighbourhood of any point $z \in \mathbb{C}$ on which the Cauchy-Riemann equations are satisfied. Therefore, f is nowhere analytic.

Exercise 1.2 (Path-dependent Derivative): Find the derivative of following functions $f(z)$ along an appropriate path:

(a) $x^2 + y^2 + 2xyi$

(b) $|z|^2 = x^2 + y^2$

(c) x^2y^2

(d) $(2x^2 + y) + i(y^2 - x)$

Also, justify that f is nowhere analytic in each case.

Answers:

(a) The Cauchy-Riemann equations hold good, only if $y = 0$, and f is differentiable along the x -axis with $f'(z) = \frac{\partial f}{\partial x} = 2x$ along it.

(b) The Cauchy-Riemann equations hold good along the imaginary axis, and $f'(z) = 3y^2 - 1$ for $y \in \mathbb{R}$

(c) The Cauchy-Riemann equations hold good along the coordinate axes, and $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 0$ along them

(d) The Cauchy-Riemann equations hold good along the straight line $y = 2x$, and $f'(z) = \frac{\partial f}{\partial x} = 4x - i$ along the path

However, in (a)-(d), $\exists$ no neighbourhood of any point $z \in \mathbb{C}$ on which the Cauchy-Riemann equations are satisfied. Therefore, f is nowhere analytic.

Definition 1.2 (Entire Function): Everywhere differentiable functions are analytic on $\mathbb{C}$, and everywhere analytic functions are differentiable on $\mathbb{C}$. A complex function $f(z)$ which is analytic throughout the complex plane $\mathbb{C}$ is called an *entire* function.

Exercise 1.3 (Entire Functions): Explain why the following functions $f(z)$ are entire, and find their derivatives at $z \in \mathbb{C}$:

(a) $x^3 - 3xy^2 + i(3x^2y - y^3)$

(b) $\sin x \cosh y + i \cos x \sinh y$.

Answers: Let $f = u + iv$. The Cauchy-Riemann equations are satisfied, and the first partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous at every point z of the complex plane. Therefore, f is everywhere differentiable and hence entire.

(a) $f'(z) = 3z^2$

(b) $f'(z) = \cos x \cosh y - i \sin x \sinh y = \cosh z$.

Exercise 1.4 (Entire and Nowhere Analytic): Explain, in detail, why

(a) $f(z) = e^x \cos y + i e^x \sin y$ is everywhere analytic

(b) $g(z) = e^x \sin y + i e^x \cos y$ is nowhere analytic.

Answers:

(a) Let $f = u + iv$. The Cauchy-Riemann equations are satisfied, and the first partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous at every point z of the complex plane. Therefore, f is entire.

(b) Let $g = p + iq$. There exists no point $z \in \mathbb{C}$, where $\frac{\partial p}{\partial x} = \frac{\partial q}{\partial y}$ and $\frac{\partial p}{\partial y} = -\frac{\partial q}{\partial x}$. Thus g is nowhere analytic.

Exercise 1.5 (Analytic on a Domain): Show that $f(z) = \frac{x^3+xy^2+x}{x^2+y^2} + i \frac{x^2y+y^3-y}{x^2+y^2}$ is analytic in an appropriate domain, and find its derivative on the domain.

Answer: Write $u(x, y) = \frac{x^3+xy^2+x}{x^2+y^2}$, $v(x, y) = \frac{x^2y+y^3-y}{x^2+y^2}$. Then u and v are continuous, provided $x^2 + y^2 \neq 0$, and the Cauchy-Riemann equations are satisfied on $\mathbb{C} - \{0\}$. Therefore, $f = u + iv$ is analytic on $\mathbb{C} - \{0\}$ with $f'(z) = 1 - 1/z^2$ for $z \neq 0$

Exercise 1.6 (Choice of the Constants): Find the constants a , b , c and d such that the following functions $f(z)$ are entire, and find $f'(z)$ in either case.

(a) $(3x + ay + 5) + i(2x + by - 3)$

(b) $e^x (\cos ay + i \sin(y + b))$

(c) $x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$

Answers:

- (a) The function f is differentiable on $\mathbb{C}$, only if $a = -2$, $b = 3$. Thus $f(z) = (3x - 2y + 5) + i(2x + 3y - 3)$ is entire and $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 3 + 2i$ at each point z .
- (b) The function f is differentiable on $\mathbb{C}$, only if $a = 1$ and $b = 0$. Thus $f(z) = e^x (\cos y + i \sin y) = e^z$ is entire and $f'(z) = e^z$ at each point z .
- (c) The Cauchy-Riemann equations hold good, only if $a = 2$, $b = -1$, $c = -1$, $d = 2$. Thus $f(z) = x^2 + 2xy - y^2 + i(-x^2 + 2xy + y^2)$ is entire and $f'(z) = \frac{\partial f}{\partial x} = 2((x + y) + i(y - x))$ at each point z .

1.2 Harmonic functions

Definition 1.3: A function $\omega(x, y)$ with continuous first and second order partial derivatives on a domain $\mathcal{D}$ is *harmonic*, if it satisfies the Laplace's equation $\nabla^2 \omega \equiv \frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} = 0$ on $\mathcal{D}$.

Theorem 1.2 (Components of an Analytic Function): Let $f(z) = u(x, y) + i v(x, y)$ be analytic on a domain $\mathcal{D}$. Then the real and imaginary parts $u \equiv u(x, y)$ and $v \equiv v(x, y)$ are harmonic on $\mathcal{D}$.

In order to prove that a function $\psi(x, y)$ is neither the real part nor the imaginary part of an analytic function f , it is enough to prove that ψ is nonharmonic.

Definition 1.4: If $f = u + i v$ is analytic, v is said to be a “harmonic conjugate of u ” or “ u is a harmonic conjugate of $-v$ ”.

To find the analytic function $f(z)$, where its real part u is given: First write the derivative formula

$$\frac{df}{dz} = \frac{\partial u}{\partial x}(x, y) - i \frac{\partial u}{\partial y}(x, y). \quad (1.3)$$

Then replace x with z and y with 0 in (1.3), and then integrate with respect to z , we get the corresponding analytic function f as a function of z . The imaginary part $v(x, y)$ of f is the harmonic conjugate of u .

To find the analytic function $f(z)$, where its imaginary part v is given: Rewrite the derivative formula as

$$\frac{df}{dz} = \frac{\partial v}{\partial y}(x, y) + i \frac{\partial v}{\partial x}(x, y). \quad (1.4)$$

Then replace x with z and y with 0 in (1.4), and then integrate with respect to z , we get the corresponding analytic function f as a function of z . The real part $u(x, y)$ of f is the harmonic conjugate of $-v$.

Exercise 1.7 (Harmonic Conjugates and Analytic Functions): Verify that the following functions $u \equiv u(x, y)$ are harmonic on the entire complex plane. Also, find an analytic function with real part u , and find the harmonic conjugate $v(x, y)$ of u in each case:

- (a) $x^3 - 3xy^2 - 5y$
- (b) $\cos x \cosh y$
- (c) $\sin x \cosh y$
- (d) $e^{x^2-y^2} \cos 2xy$
- (e) $e^{2xy} \sin(x^2 - y^2)$
- (f) $3x^2 - 3y^2 - 4xy$.

Answers:

- (a) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0$ and u is harmonic on $\mathbb{C}$. The harmonic conjugate of u is $v(x, y) = 3x^2y + 5x - y^3 + c$, and the corresponding analytic function is $f(z) = z^3 + 5iz + c_1$
- (b) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\cos x \cosh y + \cos x \cosh y = 0$ and u is harmonic on $\mathbb{C}$. The harmonic conjugate of u is $v(x, y) = -\sin x \sinh y + c$, and the corresponding analytic function is $f(z) = \cos z + c_1$
- (c) The analytic function with u as the real part is $f(z) = \sin z + ic = \sin x \cosh y + i(\cos x \sinh y + c)$. The imaginary part $v = \cos x \sinh y + c$ is the harmonic conjugate of u
- (d) The analytic function with u as the real part is $f(z) = e^{z^2} + ic = e^{x^2-y^2} \cos 2xy + i(e^{x^2-y^2} \sin 2xy + c)$. The imaginary part $v = e^{x^2-y^2} \sin 2xy + c$ is the harmonic conjugate of u .

- (e) The analytic function with u as the real part is $f(z) = i e^{-iz^2} + c$.
- (f) The analytic function with u as the real part is $f(z) = (2i + 3)z^2 + ic = 3x^2 - 3y^2 - 4xy + i(2x^2 - 2y^2 + 6xy + c)$. The imaginary part $v = 2x^2 - 2y^2 + 6xy + c$ is the harmonic conjugate of u .

Exercise 1.8: Find the analytic function $f(z) = u + iv$, where the real and imaginary parts are connected by the relation

(a) $u - v = (x - y)(x^2 + 4xy + y^2)$

(b) $u - v = \frac{\cos x + \sin x - e^{-y}}{2(\cos x - \cosh y)}$

Answers:

(a) $f(z) = c - iz^3$

(b) $f(z) = -\frac{1}{2} \cot \frac{z}{2} + c$

1.3 Orthogonality through Analytic Function

Theorem 1.3: Suppose that

$$f(z) = u(x, y) + i v(x, y) \quad (1.5)$$

is analytic at a point P . The level curves

$$u(x, y) = \alpha \text{ and } v(x, y) = \beta \quad (1.6)$$

are orthogonal at P .

Exercise 1.9 (Orthogonality through the Analyticity): Examine the orthogonality of the level curves

(a) $4x^2 + 5x - 4y^2 + 9 = a$ and $8xy + 5y - 1 = b$

(b) $\cos x \cosh y = a$ and $\sinh y \sin x = b$

by using an appropriate analytic function.

Answers:

- (a) Consider $f(z) = u + i v$, where $u = 4x^2 + 5x - 4y^2 + 9$, $v = 8xy + 5y - 1$. Then f is analytic function, and hence the level curves $u = a$ and $v = b$ are orthogonal.

- (b) Let $f(z) = u + iv$, where $u = \cos x \cosh y$, $v = -\sinh y \sin x$. Then f is analytic function and the level curves $u = a$ and $v = b$ are orthogonal.

Exercise 1.10: Using an appropriate analytic function, find the orthogonal trajectories of the family of curves:

- (a) $x^4 + y^4 - 6x^2y^2 = \text{constant}$
 (b) $3e^x \cos y - 5xy = \text{constant}$

Answers:

- (a) $x^y - xy^3 = \text{constant}$
 (b) $6e^x \sin y + 5(x^2 - y^2) = \text{constant}$

1.4 The Complex Potential

Consider a frictionless, irrotational planar flow of an incompressible fluid in a velocity field $\mathbf{V}(x, y) = f_1(x, y)\mathbf{i} + f_2(x, y)\mathbf{j}$. Then there corresponds a scalar potential $\phi(x, y)$ such that $\mathbf{V} = \text{grad } \phi$, which is harmonic. Therefore, ϕ is the real part of some analytic function, say $f(z)$.

- The potential ϕ due to the velocity field $\mathbf{V}$ is known as the velocity potential
- The analytic function $f(z)$ with ϕ as the real part is known as the complex velocity potential or the complex potential
- The imaginary part ψ of f is a harmonic conjugate of ϕ , and is the stream function of the fluid flow
- The level curves corresponding to ϕ are known as *equipotentials*, and the level curves corresponding to ψ are called the stream lines.

Given the velocity potential ϕ , to find the complex potential $f(z)$ and the velocity field $\mathbf{V}$: Replace x with z and y with 0 in the derivative formula $f'(z) = \frac{\partial \phi}{\partial x} - i \frac{\partial \phi}{\partial y}$, and integrate. Separate $f(z)$ into real and imaginary parts. The imaginary part $\psi(x, y)$ gives the stream function of the fluid flow. The corresponding velocity field is $\mathbf{V}(x, y) = \left\langle \frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right\rangle$.

Given the velocity field $\mathbf{V}(x, y) = \langle f_1(x, y), f_2(x, y) \rangle$, to find the velocity potential ϕ and the complex potential $f(z)$: Determine the scalar potential $\phi(x, y)$ using the irrotationality of $\mathbf{V}$, which is the velocity potential of the flow. Then adopt the previous procedure.

Exercise 1.11 (Complex Potentials): Show that each of the following functions $\varphi(x, y)$ represents the *velocity potential* of a fluid flow. Also, find the complex potential ω , the stream function $\psi(x, y)$ and the corresponding *velocity field* $\vec{v}$:

- (a) $e^{e^x \cos y} \cos(e^x \sin y)$
- (b) $4x^2 + 5x - 4y^2$
- (c) $\sinh x \cos y$
- (d) $\frac{5}{2}(x^2 - y^2)$
- (e) $\ln(x^2 + y^2)$.

Answers:

- (a) φ is harmonic, and hence represents the *velocity potential* of a fluid flow. The complex potential of the flow is $\omega(z) = e^{e^z} + i c$. Separating this into real and imaginary parts, we see that $\omega = e^{e^x \cos y} \cos(e^x \sin y) + i e^{e^x \cos y} \sin(e^x \sin y)$. Note that the real part of ω is φ , whereas its imaginary part $\psi = e^{e^x \cos y} \sin(e^x \sin y) + c$ represents the stream function of the fluid flow
- (b) φ is harmonic, and hence represents the *velocity potential* of a fluid flow. The complex potential of the flow is $\omega(z) = 4z^2 + 5z + i k$. Separating this into real and imaginary parts, we see that $\omega = 4x^2 + 5x - 4y^2 + i(8xy + 5y + k)$. Note that the real part of ω is φ , whereas its imaginary part $\psi = 8xy + 5y + k$ represents the stream function of the fluid flow
- (c) The complex potential of the flow is $f(z) = \sinh z + i c$, and $\psi(x, y) = \cosh x \sin y + c$ is the stream function
- (d) The complex potential of the flow is $f(z) = \frac{5}{2}z^2 + c$, and the stream function is $\psi(x, y) = 5xy + i c$
- (e) The complex potential of the flow is $f(z) = 2 \ln z + c$, and the stream function is $\psi(x, y) = \tan^{-1} \frac{y}{x}$.

The corresponding velocity field in each case is given by $\vec{v} = \left\langle \frac{\partial \varphi}{\partial x}, \frac{\partial \varphi}{\partial y} \right\rangle$.

Theorem 1.4 (Cauchy-Reimann Equations in the Polar Form): *Let*

$$f(z) = f(re^{i\theta}) = u(r, \theta) + i v(r, \theta) \quad (1.7)$$

be analytic at $z = re^{i\theta}$. Then

(a) the partial derivatives $\frac{\partial u}{\partial r}$, $\frac{\partial u}{\partial \theta}$, $\frac{\partial v}{\partial r}$ and $\frac{\partial v}{\partial \theta}$ are continuous, and

(b) satisfy

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \text{ and } \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta} \quad (1.8)$$

at every point of some neighborhood of z . Then f is analytic at z , and the derivative of f is

$$f'(z) = e^{-i\theta} \frac{\partial f}{\partial r} = e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right). \quad (1.9)$$

Exercise 1.12 (Cauchy-Riemann Equations in the Polar Form): Let f be the principal square root function: $f(z) = z^{1/2} = r^{1/2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$, defined on $\mathcal{D} = \{z = r e^{i\theta} : r > 0, -\pi < \theta < \pi\}$. Verify that f satisfies the Cauchy-Riemann equations, and find its derivative at any point of $\mathcal{D}$.

Answer: Write $u = r^{1/2} \cos \frac{\theta}{2}$ and $v = r^{1/2} \sin \frac{\theta}{2}$. Then

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{2r^{1/2}} \cos \frac{\theta}{2} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \\ \frac{\partial u}{\partial \theta} &= -\frac{1}{2} r^{1/2} \sin \frac{\theta}{2} = -\frac{1}{r} \frac{\partial v}{\partial r}. \end{aligned}$$

Thus the Cauchy-Riemann equations hold good at every point $z \in \mathcal{D}$. Also, the partial derivatives $\frac{\partial u}{\partial r}$, $\frac{\partial u}{\partial \theta}$, $\frac{\partial v}{\partial r}$ and $\frac{\partial v}{\partial \theta}$ are continuous on $\mathcal{D}$. Therefore, f is analytic on $\mathcal{D}$, and its derivative is $f'(z) = e^{-i\theta} \frac{\partial f}{\partial r} = 1/2r^{1/2} e^{i\theta/2} = 1/2z^{1/2}$.

Exercise 1.13 (Equipotentials and Stream Lines): Let $f(z) = z^{1/3} = r^{1/3} \left(\cos \frac{\theta}{3} + i \sin \frac{\theta}{3} \right)$, defined on $\mathcal{D} = \{z = r e^{i\theta} : r > 0, -\pi < \theta < \pi\}$. Verify that f is analytic, and find its derivative at any point of $\mathcal{D}$. Further, if f represents the complex potential of an electrostatic charge flow, find the equipotentials and the lines of force.

Answer: Write $u = r^{1/3} \cos \frac{\theta}{3}$ and $v = r^{1/3} \sin \frac{\theta}{3}$. Then

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{1}{3} \cdot r^{-2/3} \cos \frac{\theta}{3}, \quad \frac{\partial u}{\partial \theta} = -\frac{1}{3} r^{1/3} \sin \frac{\theta}{3}, \\ \frac{\partial v}{\partial r} &= \frac{1}{3} \cdot r^{-2/3} \sin \frac{\theta}{3}, \quad \frac{\partial v}{\partial \theta} = \frac{1}{3} r^{1/3} \cos \frac{\theta}{3}. \end{aligned}$$

Thus $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$. That is, the Cauchy-Riemann equations hold good at every point $z \in \mathcal{D}$. Also, the partial derivatives $\frac{\partial u}{\partial r}$, $\frac{\partial u}{\partial \theta}$, $\frac{\partial v}{\partial r}$ and $\frac{\partial v}{\partial \theta}$ are continuous on $\mathcal{D}$. Therefore, f is

analytic on $\mathcal{D}$, and its derivative is

$$\begin{aligned} f'(z) &= e^{-i\theta} \frac{\partial f}{\partial r} = e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = e^{-i\theta} \frac{1}{3} \cdot r^{-2/3} \left(\cos \frac{\theta}{3} + i \sin \frac{\theta}{3} \right) \\ &= \frac{1}{3} \cdot r^{-2/3} e^{-i\theta} e^{i\theta/3} = \frac{1}{3} \cdot r^{-2/3} e^{-2i\theta/3} = 1/3 z^{2/3}. \end{aligned}$$

The real and imaginary parts of f represent the velocity potential, and the stream function respectively. The corresponding level curves are the equipotentials and the lines of force respectively, and are given by $r^{1/3} \cos \frac{\theta}{3} = \alpha$ and $r^{1/3} \sin \frac{\theta}{3} = \beta$ respectively.

■ Glossary of Terms

Analytic Function: Let $\mathcal{D} \subset \mathbb{C}$ and $z_0 \in \mathcal{D}$. A function $f : \mathcal{D} \rightarrow \mathbb{C}$ is said to be analytic at $z_0 \in \mathcal{D}$, if it is differentiable on some neighborhood of z_0 .

Cauchy-Riemann Equations (Cartesian Form): Let $f(z) = u(x, y) + i v(x, y)$ be analytic at a point $z = (x, y)$. Then $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ hold good at z .

Cauchy-Riemann Equations (Polar Form): Let $f(z) = f(re^{i\theta}) = u(r, \theta) + i v(r, \theta)$ be analytic at $z = re^{i\theta}$. Then $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$.

Complex Potential: The analytic function $f(z) = u(x, y) + i v(x, y)$ which completely describes the fluid flow pattern is called the complex potential.

Derivative Formula (Cartesian Form): Let $f(z) = u(x, y) + i v(x, y)$ be analytic at a point $z = (x, y)$. Then $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$.

Derivative Formula (Polar Form): Let $f(z) = f(re^{i\theta}) = u(r, \theta) + i v(r, \theta)$ be analytic at $z = re^{i\theta}$. Then $f'(z) = e^{-i\theta} \frac{\partial f}{\partial r} = e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right)$.

Entire Function: A complex function which is analytic at every point of the complex plane $\mathbb{C}$

Equipotentials: Let $f(z) = \varphi(x, y) + i \psi(x, y)$ be the complex potential of a fluid flow. Then paths along which a constant velocity potential is observed are the equipotentials.

Harmonic Conjugates: Let $f(z) = u(x, y) + i v(x, y)$ be analytic at a point $z = (x, y)$. Then the imaginary part $v(x, y)$ is a harmonic conjugate of the real part $u(x, y)$.

Harmonic Function: A function $\omega(x, y)$ with continuous first and second order partial derivatives on a domain $\mathcal{D}$ is *harmonic*, if it satisfies the Laplace's equation $\nabla^2 \omega \equiv \frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} = 0$ on $\mathcal{D}$. If $f(z) = u(x, y) + i v(x, y)$ is analytic, the real and imaginary parts u and v are harmonic.

Nowhere Analytic Function: For every point $z \in \mathbb{C}$, if there corresponds no neighborhood on which f is differentiable, then f is nowhere analytic.

Stream Function: The imaginary part of the complex potential of a fluid flow.

Streamlines: Let $f(z) = \varphi(x, y) + i\psi(x, y)$ be the complex potential of a fluid flow. Then the paths of motion of the fluid particles are the streamlines of the flow.

Velocity Potential: The real part of the complex potential of a fluid flow.

Module 2

Conformal Mapping

Definition 2.1 (Conformal Mapping): A complex mapping is said to be *conformal* at a point z_0 of its domain, provided it preserves the angles at z_0 .

Theorem 2.1 (Conformality in terms of Analyticity): A mapping

$$w = f(z) = u(x, y) + i v(x, y) \quad (2.1)$$

is conformal at a point $z_0 \in D$ if and only if it is analytic at z_0 and

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \neq 0 \text{ at } z_0. \quad (2.2)$$

The angle of rotation is $\alpha = \text{Arg } f'(z_0)$ and the scale factor is

$$J = \left(\frac{u, v}{x, y} \right) = |f'(z_0)|^2 \neq 0. \quad (2.3)$$

Definition 2.2 (The Mapping e^z): If $z = x + iy$, the complex exponential mapping is defined by

$$\omega = e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y) = u + iv. \quad (2.4)$$

The real and imaginary parts of (2.4) are

$$u = u(x, y) = e^x \cos y \text{ and } v = v(x, y) = e^x \sin y. \quad (2.5)$$

If $\omega = \rho e^{i\psi} = e^z$, then $(\cos y + i \sin y) = \rho e^{i\psi} = e^x e^{iy}$ so that $\rho = |\omega| = e^x > 0$ and $\psi = y$ are the modulus and argument of the exponential map (2.4), where the principal value of $\psi = y$ lies in $(0, 2\pi]$.

Exercise 2.1 (The Exponential Mapping): Determine the conformality of $f(z) = e^z$ and the orthogonality of the level curves corresponding to its real and imaginary parts. Also, express it in the polar form.

Exercise 2.2 (Images under the Exponential Mapping): Find the images of the following curves/regions under $f(z) = e^z$:

- (a) the coordinate axes
- (b) the curves $e^x \cos y = a$ and $e^x \sin y = b$
- (c) the straight line $y = x$
- (d) the rectangular region enclosed by edges $x = \pm 1, y = 0, y = 2\pi$
- (e) the infinite horizontal strip $-\infty < x < \infty, 0 \leq y \leq \pi/4$.

Answers:

- (a) The image of the real axis is the positive real axis $u = e^x > 0$, $v = 0$ in the uv -plane, and the image of the imaginary axis is the unit circle $u^2 + v^2 = 1$ with centre at $(0, 0)$
- (b) The straight lines $u = a$ and $v = b$ respectively
- (c) The *logarithmic spiral* $\rho = e^\psi$
- (d) The ring shaped region, centred at 0, enclosed by the circles with inner radius $1/e$ and outer radius e
- (e) The unbounded region in the first quadrant enclosed by the infinite rays $\psi = 0$ and $\psi = \pi/4$.

Definition 2.3 (The Mapping $f(z) = z^2$): The mapping z -squared is expressed in the rectangular coordinates as

$$w = f(z) = u + iv = z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy). \quad (2.6)$$

The real and imaginary parts of f are

$$u(x, y) = x^2 - y^2 \text{ and } v(x, y) = 2xy. \quad (2.7)$$

The *level curves* $u = x^2 - y^2 = a$ and $v = 2xy = b$ are families of orthogonal trajectories, which are *rectangular hyperbolas*.

If $w = z^2$ is the complex potential of a fluid-flow, then the real part $u = x^2 - y^2$ represents the *velocity potential* and the level curves $u \equiv x^2 - y^2 = a$ are *equipotentials*. Whereas, the imaginary part $v = 2xy$ denotes the *stream function*, and the level curves $v \equiv 2xy = b$ are *lines of force* or *stream lines* of the fluid-flow.

Exercise 2.3 (The Mapping z -squared): Determine the conformality of $f(z) = z^2$ and the orthogonality of the level curves corresponding to its real and imaginary parts. Also, express it in the polar form.

Exercise 2.4 (Images under the Mapping z -squared): Find the images of the following curves/regions under $f(z) = z^2$:

- (a) the coordinate axes
- (b) the rectangular parabolas $x^2 - y^2 = 4$ and $xy = 16$
- (c) the straight line $y = x$
- (d) the vertical line $x = 1$
- (e) the horizontal line $y = 1$

- (f) the upper half of the xy -plane
- (g) the first quadrant $\{(r, \theta) : r \geq 0, 0 \leq \theta \leq \pi/2\}$
- (h) the image of the circle with centre at the origin and radius a
- (i) The annular sector $z = re^{i\theta}$, $a \leq r \leq b$ and $0 \leq \theta \leq \pi/3$
- (j) The region enclosed by the square $\{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$.

Answers:

- (a) The image of the real axis is the positive real axis $u > 0, v = 0$, and the image of the imaginary axis is the nonpositive portion of the real axis $u \leq 0, v = 0$
- (b) the straight lines $u = 4$ and $v = 32$
- (c) the nonnegative imaginary axis in the uv -plane
- (d) the negatively oriented parabola $v^2 = -4(u - 1)$ with vertex at $(1, 0)$, and focus at the origin $(0, 0)$
- (e) the positively oriented parabola $v^2 = 4(u + 1)$ with vertex at $(-1, 0)$, and focus at the origin $(0, 0)$
- (f) the entire uv -plane
- (g) the upper half uv -plane $\{(\rho, \psi) : \rho \geq 0, 0 \leq \psi \leq \pi\}$
- (h) the enlarged circle with centre at the origin and radius a^2 in the uv -plane
- (i) The enlarged annular sector $z = \rho e^{i\psi}$, $a^2 \leq \rho \leq b^2$ and $0 \leq \psi \leq 2\pi/3$
- (j) The region enclosed by the parabolas $v^2 = -4(u - 1)$ and $v^2 = 4(u + 1)$, which lies above the uv -plane.

Exercise 2.5 (Conformality): Find the regions on which the following functions are conformal:

- (a) $f(z) = z^2 + 2iz - 3$
- (b) $f(z) = z^3 - 5z + 1$
- (c) $f(z) = \sin z$
- (d) $f(z) = \cosh z$
- (e) $f(z) = ze^{z^2-2}$.

Answers:

- (a) Note that $f'(z) = 2z + 2i = 0$ only if $z = i$. Therefore, f is conformal on $\mathbb{C} - \{i\}$
- (b) Since $f(z) = 3z^2 - 5 = 0$ implies that $z = \pm\sqrt{\frac{5}{3}}$, f is conformal on $\mathbb{C} - \left\{-\sqrt{\frac{5}{3}}, \sqrt{\frac{5}{3}}\right\}$

- (c) Since $f'(z) = \cos z = 0$ only if $z_n = (2n + 1)\pi/2$ for $n = 0, \pm 1, \pm 2, \dots$, f is conformal on $\mathbb{C} - \{(2n + 1)\pi/2 : n \in \mathbb{Z}\}$
- (d) Since $f'(z) = \sinh z = 0$ only if $z = 0$, f is conformal on $\mathbb{C} - \{0\}$
- (e) f is conformal on $\mathbb{C} - \{\pm i\sqrt{2}/2\}$.

2.1 Bilinear Transformations

Definition 2.4: Let $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. The mapping

$$f(z) = \frac{az + b}{cz + d} \quad (2.8)$$

is called a *bilinear* or *linear fractional* transformation, with determinant $ad - bc \neq 0$.

If $c \neq 0$, f is conformal on the domain $\mathbb{C} - \{-d/c\}$ with $ad \neq bc$.

Exercise 2.6 (Invariant Points): The invariant points or fixed points of a complex transformation $f(z)$ are the roots of the equation $f(z) = z$. Find the invariant points of the bilinear transformations

- (a) $f(z) = \frac{4z+1}{2z+3}$
- (b) $f(z) = \frac{z-1}{z+1}$
- (c) $f(z) = \frac{6z-9}{z}$.

Answers:

- (a) The invariant points are $z = 1$ and $-1/2$
- (b) The invariant points are $z = \pm i$
- (c) $z = 3$ is the only invariant point

Theorem 2.2 (Implicit Formula for a Bilinear transformation): If $w = f(z)$ is a unique bilinear transformation that maps three distinct points z_1, z_2 and z_3 onto three points w_1, w_2 and w_3 respectively, then the cross-ratio is invariant under f , that is

$$\frac{w - w_1}{w - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1} = \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1}. \quad (2.9)$$

Exercise 2.7 (Bilinear Transformations): Find the bilinear transformation $w = f(z)$ and its invariant points, that sends

- (a) $1, i, -1$
- (b) $-1, 0, 2$
- (c) $1, 2, 3$

onto $0, 1, \infty$ respectively.

Answers: Use the formula $w = \frac{z-z_1}{z-z_3} \frac{z_2-z_3}{z_2-z_1}$.

(a) $w = \frac{i-iz}{z+1}$, the invariant points of w are $\frac{\sqrt{3}-1}{2}(1+i)$, $-\frac{\sqrt{3}+1}{2}(1+i)$

(b) $w = \frac{2z+2}{2-z}$, invariant points of w are $\pm\sqrt{2}i$

(c) $f(z) = w = \frac{1-z}{z-3}$, invariant points of w are $1 \pm \sqrt{2}$.

Definition 2.5 (Cross-Ratio): The image of z under the bilinear map (2.9), which sends the points z_1, z_2 and z_3 onto $w_1 = 0, w_2 = 1$ and $w_3 = \infty$ respectively, is called the *cross-ratio* of the points z, z_1, z_2 and z_3 taken in order.

The cross-ratio of four points z_1, z_2, z_3 and z_4 taken in order is given by $[z_1, z_2, z_3, z_4] = \frac{z_1 - z_2}{z_1 - z_4} \frac{z_3 - z_4}{z_3 - z_2}$.

Exercise 2.8 (Cross-ratio): Evaluate the following cross-ratios:

(a) $[0, 1, i, -1]$

(b) $[0, 1, i, \infty]$

(c) $[i-1, \infty, 1+i, 0]$

Answers:

(a) i

(b) $(1+i)/2$

(c) i

Exercise 2.9 (Bilinear Transformations): Find the bilinear transformation $w = f(z)$ that sends

(a) $0, 1, \infty$ onto $0, i, 2$

(b) $-1, 0, 1$ onto $i, \infty, 0$

(c) $-i, 1, i$ onto $-1, 0, 1$

(d) $0, i, -i$ onto $-1, 1, 0$

(e) $0, -i, -1$ onto $i, 1, 0$

Answers:

(a) $z = \frac{w-0}{w-2} \cdot \frac{i-2}{i-0}$ or $w = \frac{2z}{z-2i-1}$

$$(b) \frac{z+1}{z-1} \cdot \frac{0-1}{0+1} = \frac{w-i}{w-0} \text{ or } w = \frac{i(z-1)}{2z}$$

$$(c) w = \frac{1-z}{1+z}$$

$$(d) w = \frac{z+i}{3z-i}$$

$$(e) w = \frac{i(z+1)}{1-z}$$

2.2 Images under Bilinear Transformations

Exercise 2.10 (Images under Bilinear Transformations): Find the images of the regions $\mathcal{A} : |z| \leq 1$, $\mathcal{B} : |z - i| \leq 1$, $\mathcal{C} : x \geq 0$ and $\mathcal{D} : y \leq 1$ under the following transformations $w = f(z)$:

$$(a) \frac{i}{z}$$

$$(b) \frac{2}{z-i}$$

$$(c) \frac{z+i}{z-i}$$

$$(d) \frac{z-1}{z}.$$

Answers:

- (a) The image of $\mathcal{A}$ is the outer part of the unit circle $|\omega| = 1$, the image of $\mathcal{B}$ is the right side of the plane $u = 1/2$, the image of $\mathcal{C}$ is the upper half uv -plane, the image of $\mathcal{D}$ is the exterior of the circle $|\omega - 1/2| = 1/2$ in the uv -plane
- (b) The image of $\mathcal{A}$ is the outer part of the upper portion of the line $v = 1$, the image of $\mathcal{B}$ is the exterior of the circle $|\omega| = 2$, the image of $\mathcal{C}$ is the right half of the uv -plane, the image of $\mathcal{D}$ is the upper-half uv -plane
- (c) The image of $\mathcal{A}$ is the left-half uv -plane, the image of $\mathcal{B}$ is the exterior of the circle $|\omega - 1| = 2$, the image of $\mathcal{C}$ is the upper half uv -plane, the image of $\mathcal{D}$ is the portion of the uv -plane to the left of the line $u = 1$
- (d) The image of $\mathcal{A}$ is the exterior of the circle $|\omega - 1| = 1$, the image of $\mathcal{B}$ is the upper portion of the line $v = u$ in the uv -plane, the image of $\mathcal{C}$ is the right portion of the line $u = 1$ in the uv -plane, the image of $\mathcal{D}$ is the exterior of the circle with centre $(1, 1/2)$ and radius $1/2$ in the uv -plane.

Exercise 2.11 (Images under a Bilinear Transformation): Find the images of the following curves/regions under the given bilinear transformation:

$$(a) \text{ The disk } \mathcal{A} : |z| \leq 2, \text{ the interior } \mathcal{B} \text{ of the circle } |z| = 1; f(z) = \frac{z+2}{z-1}$$

$$(b) \text{ The interior of the circle } |z| = 1; f(z) = z/(1 - z).$$

Answers:

- (a) The exterior of the circle $|z - 2| = 2$, The left plane $u < -1/2$
- (b) The right side portion of the line $u = -1/2$ in the uv -plane.

Glossary of Terms

Bilinear Transformation: Let $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. The mapping $f(z) = \frac{az+b}{cz+d}$ is called a bilinear transformation, with determinant $ad - bc \neq 0$.

Complex Exponential Mapping: For all $z \in \mathbb{C}$, we have $e^z = e^{x+iy} = e^x \cos y + i e^x \sin y$. If $\omega = \rho e^{i\psi} = e^z$, then $\rho = |\omega| = e^x > 0$ and $\psi = y$ are the modulus and argument of the exponential map respectively, where the principal value of $\psi = y$ lies in $(0, 2\pi]$.

Complex Squared Mapping: For all $z \in \mathbb{C}$, we have $z^2 = (x^2 - y^2) + i(2xy)$. If $z = r e^{i\theta}$ and $\omega = \rho e^{i\psi} = z^2$, then $\rho = |\omega| = r^2 \geq 0$ and $\psi = 2\theta$ are the modulus and argument of the squared map respectively.

Conformal Mapping: A complex mapping is said to be *conformal* at a point z_0 of its domain, provided it preserves the angles at z_0 .

Cross-Ratio: The image of z under the bilinear map (2.9), which sends the points z_1, z_2 and z_3 onto $w_1 = 0, w_2 = 1$ and $w_3 = \infty$ respectively, is called the *cross-ratio* of the points z, z_1, z_2 and z_3 taken in order. The cross-ratio of four points z_1, z_2, z_3 and z_4 taken in order is given by $[z_1, z_2, z_3, z_4] = \frac{z_1 - z_2}{z_1 - z_4} \cdot \frac{z_3 - z_4}{z_3 - z_2}$.

Implicit Formula for a Bilinear transformation: The bilinear transformation that maps three distinct points z_1, z_2 and z_3 onto three points w_1, w_2 and w_3 respectively is given by $\frac{w-w_1}{w-w_3} \cdot \frac{w_2-w_3}{w_2-w_1} = \frac{z-z_1}{z-z_3} \cdot \frac{z_2-z_3}{z_2-z_1}$.

Invariant Points: The roots of the equation $f(z) = z$ are called the invariant points or fixed points of a complex transformation $f(z)$.

Module 3

Contour Integration

3.1 Taylor's Series

Theorem 3.1 (Taylor's Series about a Point): Let $f(z)$ be analytic on $\mathcal{D} \subset \mathbb{C}$ and $z_0 \in \mathcal{D}$. Then the Taylor's series of f about z_0 is given by

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots, \quad (3.1)$$

where

$$a_0 = f(z_0), \quad a_n = \frac{f^{(n)}(z_0)}{n!}, \quad n \geq 1 \quad (3.2)$$

are called the coefficients of the Taylor's series (3.1).

To find the Taylor's series of a rational polynomial function f about the origin, first reduce it into partial fractions, and then employ the binomial expansion formula:

$$(1 + \eta)^\alpha = 1 + \alpha\eta + \frac{\alpha(\alpha - 1)\eta^2}{2!} + \frac{\alpha(\alpha - 1)(\alpha - 2)\eta^3}{3!} + \cdots \text{ for } |\eta| < 1, \alpha \in \mathbb{R} \quad (3.3)$$

to each partial fraction, and then simplify.

Exercise 3.1 (Taylor's Series about the Origin): Expand the following functions $f(z)$ in Taylor's series about the origin, and determine its region $\mathcal{D}$ of validity in each case:

(a) $\frac{1}{(z - 1)(z + 2)}$

(b) $\frac{1}{(z + 1)(z - 3)}$

(c) $\frac{1}{z^2 + z + 1}$

(d) $\frac{1}{(z + 2)^2(z - 3)}$

(e) $\frac{z - 2}{2z + 3}$.

Answers:

(a) $\frac{1}{(z - 1)(z + 2)} = \frac{1}{3} \sum_{n=0}^{\infty} \left(-1 + \left(-\frac{1}{2} \right)^{n+1} \right) z^n$, valid on $|z| < 1$

(b) $\frac{1}{(z + 1)(z - 3)} = \sum_{n=0}^{\infty} \left(2(-1)^n + \left(-\frac{1}{3} \right)^{n+1} \right) z^n$, valid on $|z| < 3$

(c) $\frac{1}{z^2 + z + 1} = \frac{1}{a - b} \sum_{n=0}^{\infty} \left(\frac{1}{b^{n+1}} - \frac{1}{a^{n+1}} \right) z^n$, where $a = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $b = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

(d) $\frac{1}{(z + 2)^2(z - 3)} = \frac{1}{50} \sum_{n=0}^{\infty} (-1)^{n+1} (z/2)^n + \frac{1}{20} \sum_{n=0}^{\infty} (n + 1)(-1)^{n+1} (z/2)^n - \frac{1}{75} \sum_{n=0}^{\infty} (z/3)^n$

$$(e) \frac{z-2}{2z+2} = \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{3}\right)^n z^{n+1} + \sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^{n+1} z^n.$$

Exercise 3.2 (Taylor's Series about a Point): Find the Taylor series about z_0 for the following functions $f(z)$, and mention the region $\mathcal{D}$ of validity in each case:

- (a) $1/z^2$, $z_0 = i$
- (b) $\frac{1}{(z-1)(z-3)}$, $z_0 = 4$
- (c) $\frac{1+z}{(1-z)^2}$, $z_0 = i$
- (d) $\frac{z-1}{z^2}$, $z_0 = 1$
- (e) $\frac{z-1}{3-z}$, $z_0 = 1$.

Answers: To find the Taylor's series of a rational polynomial function f about $z_0 \neq 0$:

✎ Substitute $z - z_0 = u$ in f to get a rational polynomial function $\psi(u)$.

✎ Reduce ψ into partial fractions, and then employ the binomial expansion formula (3.3) to each partial fraction, and then simplify in terms of $u = z - z_0$.

- (a) $\frac{1}{z^2} = -\sum_{n=0}^{\infty} (n+1)(i)^n (z-i)^n$, $\mathcal{D} = \{z \in \mathbb{C} : |z-i| < 1\}$
- (b) $\frac{1}{(z-1)(z-3)} = \frac{1}{2} \sum_{n=0}^{\infty} \left(1 - \frac{1}{3^{n+1}}\right) (-1)^n (z-4)^n$, $\mathcal{D} = \{z \in \mathbb{C} : |z-4| < 1\}$, which is the interior of the circle with centre at 4 and radius 1
- (c) $\frac{1+z}{(1-z)^2} = -1 + \sum_{n=0}^{\infty} \frac{2}{(1-i)^{n+1}} (z-i)^n$
- (d) $\frac{z-1}{z^2} = \sum_{n=0}^{\infty} (-1)^n (z-i)^{n+1}$
- (e) $\frac{z-1}{3-z} = \sum_{n=0}^{\infty} \frac{1}{2^n} (z-1)^n$

3.2 Laurent's Series about an Isolated Singularity

Definition 3.1 (Isolated singularity): If $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$, then z_0 is called an *isolated singularity* of f .

Consider a rational function $f(z) = \phi(z)/\psi(z)$. If $\psi(z_0) = 0$ and $\phi(z_0) \neq 0$, then z_0 is an isolated singularity of f .

Theorem 3.2 (Laurent's series): Let z_0 be an isolated singularity of f . Suppose that f be analytic on the annulus $\mathcal{A} : 0 < r < |z - z_0| < R$. The Laurent's series of f about z_0 is given by

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}}_{\text{Principal Part}} + \underbrace{\sum_{n=0}^{\infty} a_n(z - z_0)^n}_{\text{Analytic Part}}, \quad (3.4)$$

where a'_n s are called its coefficients.

Definition 3.2 (Residue at an Isolated Singularity): The coefficient a_{-1} of the first negative power $1/(z - z_0)$ in (3.5) is known as the *residue* $\text{Res}(f; z_0)$ of f at $z = z_0$.

The residue should be computed at an isolated singularity, which lies within the region of validity of the Laurent's series. The residue is not defined at an isolated singularity not lying in the validity region.

Exercise 3.3: Find a Laurent's series of each of the following functions $f(z)$ on the domains of validity mentioned, and find the residues at its isolated singularities where ever possible:

- (a) $1/z^2(1 - z)$; $\mathcal{R}_1 : 0 < |z| < 1$, $\mathcal{R}_2 : |z| > 1$, $\mathcal{R}_3 : 0 < |z - 1| < 1$ and $\mathcal{R}_4 : |z - 1| > 1$
- (b) $1/(z + 1)(z - 2)$, $\mathcal{R}_1 : 0 < |z + 1| < 3$, $\mathcal{R}_2 : |z + 1| > 3$, $\mathcal{R}_3 : 0 < |z - 2| < 3$ and $\mathcal{R}_4 : |z - 2| > 3$
- (c) $\frac{1}{z(z + 5)}$; $\mathcal{R}_1 : 0 < |z| < 5$, $\mathcal{R}_2 : |z| > 5$, $\mathcal{R}_3 : 0 < |z + 5| < 5$, and $\mathcal{R}_4 : |z + 5| > 5$
- (d) $\frac{1}{z^2 + 1}$; $\mathcal{R}_1 : 0 < |z| < 1$, $\mathcal{R}_2 : |z| > 1$
- (e) $-\frac{5}{2z} + \frac{2}{z^2} + \frac{3}{z + 1} - \frac{1}{2} \frac{1}{z + 2}$; $\mathcal{R}_1 : 0 < |z| < 1$, $\mathcal{R}_2 : 1 < |z| < 2$.

Answers:

(a) Note that $z = 0$ and $z = 1$ are isolated singularities of f .

- On $\mathcal{R}_1 : 0 < |z| < 1$: $f(z) = \sum_{n=0}^{\infty} z^{n-2} = \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \dots$, $\text{Res}(f; 0) = a_1 = 1$
- On $\mathcal{R}_2 : |z| > 1$: $f(z) = -\left\{ \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \dots \right\}$
- On $\mathcal{R}_3 : 0 < |z - 1| < 1$: $f(z) = -\sum_{n=0}^{\infty} (n + 1)(-1)^n u^{n-1} = -\sum_{n=0}^{\infty} (n + 1)(-1)^n (z - 1)^{n-1}$, $\text{Res}(f; 1) = -1$
- On $\mathcal{R}_4 : |u| > 1$: $f(z) = -\frac{1}{u^3} + \frac{2}{u^4} - \frac{3}{u^5} + \dots = -\frac{1}{(z - 1)^3} + \frac{2}{(z - 1)^4} - \frac{3}{(z - 1)^5} + \dots$

(b) Note that $z = -1$ and $z = 2$ are isolated singularities of f .

- On $\mathcal{R}_1 : 0 < |z + 1| < 3$: $f(z) = -\frac{1}{3} \left\{ \frac{1}{z + 1} + \frac{1}{3} + \frac{z + 1}{3^2} + \dots \right\}$, $\text{Res}(f; -1) = a_1 = -1/3$

- On $\mathcal{R}_2 : |z+1| > 3$: $f(z) = \frac{1}{(z+1)^3} + \frac{3}{(z+1)^4} + \frac{3^2}{(z+1)^5} + \dots$
- On $\mathcal{R}_3 : 0 < |z-2| < 3$: $f(z) = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n (z-2)^{n-1}$, $\text{Res}(f; 2) = 1$
- On $\mathcal{R}_4 : |z-2| > 1$: $f(z) = \frac{1}{(z-2)^3} - \frac{3}{(z-2)^4} - \frac{3^2}{(z-2)^5} + \dots$

(c) $z = 0$ and $z = -5$ are isolated singularities of f .

- On $\mathcal{R}_1 : 0 < |z| < 5$: $f(z) = -\frac{1}{5z} \left\{ 1 - \frac{z}{5} + \frac{z^2}{5^2} - \frac{z^3}{5^3} + \dots \right\}$, $\text{Res}(f; 0) = a_1 = -1/5$
- On $\mathcal{R}_2 : |z| > 5$: $f(z) = \frac{1}{z^2} - \frac{5}{z^3} + \frac{5^2}{z^4} - \frac{5^3}{z^5} + \dots$
- On $\mathcal{R}_3 : 0 < |z+5| < 1$: $f(z) = -\sum_{n=0}^{\infty} \frac{1}{5^{n+1}} (z+5)^{n-1}$, $\text{Res}(f; -5) = -1/5$
- On $\mathcal{R}_4 : |z+5| > 1$: $f(z) = \sum_{n=0}^{\infty} 5^n / (z+5)^{n+2}$

(d) $z = 0$ and $z = -\pm i$ are the isolated singularities of f .

- On $\mathcal{R}_1 : 0 < |z| < 1$: $f(z) = \sum_{n=0}^{\infty} (-1)^n z^{2n-1}$, $\text{Res}(f; 0) = a_1 = 1$
- On $\mathcal{R}_2 : |z| > 1$: $f(z) = \sum_{n=0}^{\infty} (-1)^n / z^{2n+3}$

(e) $z = 0$ and $z = -1, z = -2$ are the isolated singularities of f .

- On $\mathcal{R}_1 : 0 < |z| < 1$: $f(z) = -\frac{5}{2} \frac{1}{z} + \frac{2}{z^2} + \sum_{n=0}^{\infty} (-1)^n (3 - 1/2^{n+2}) z^n$
- On $\mathcal{R}_2 : 1 < |z| < 2$: $f(z) = \frac{1}{2} \frac{1}{z} - \frac{1}{z^2} + \sum_{n=2}^{\infty} 3(-1)^n / z^{n+1} - \sum_{n=0}^{\infty} (-1)^n z^n / 2^{n+2}$.

Exercise 3.4: Find the Laurent's series of each of the following functions $f(z)$ about z_0 , which converges on the given annulus $\mathcal{A}$:

- (a) $1/z(1-z)$, $z_0 = 2$, $\mathcal{A} : 1 < |z-2| < 2$
- (b) $\frac{1}{z(z+2)}$, $z_0 = -1$, $\mathcal{A} : 1 < |z-1| < 3$
- (c) $\frac{z^2}{z^2-3z+2}$, $z_0 = 3$, $\mathcal{A} : 1 < |z-3| < 2$
- (d) $\frac{z}{(z-1)(z-3)}$, $z_0 = 0$, $\mathcal{A} : 1 < |z| < 3$
- (e) $\frac{z-1}{z^2}$, $z_0 = 2$, $\mathcal{A} : 1 < |z-2| < 2$.

Answers:

- (a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} (z-2)^n + \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(z-2)^{n+1}}$

- (b) $\frac{1}{2} \sum_{n=0}^{\infty} (-1)^n / (z+1)^{n+1} + \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{3^{n+1}} (z+1)^n$
- (c) $\frac{1}{2} + 4 \sum_{n=1}^{\infty} (-1)^{n+1} / (z-3)^n - \sum_{n=1}^{\infty} \frac{(-1)^n}{2^{n+1}} (z-3)^n$
- (d) $-\frac{1}{2} \sum_{n=0}^{\infty} z^n / 3^{n+1} + \frac{1}{2} \sum_{n=0}^{\infty} 1/z^n$
- (e) $\frac{1}{4} \sum_{n=0}^{\infty} (-1)^n (n+1) (z-2)^{n+1} / 2^n + \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n (n+1) (z-2)^n / 2^n$

3.3 Poles and Residues

Let $z_0 \in \mathbb{C}$ an isolated singularity¹ of $f(z)$, and $f(z)$ be analytic on the annulus $\mathcal{A} : 0 < r < |z - z_0| < R$. Then the Laurent's series of f about z_0 is given by

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}}_{\text{Principal Part}} + \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{Analytic Part}}. \quad (3.5)$$

Definition 3.3 (Residue at an Isolated Singularity): The coefficient a_{-1} of the first negative power $1/(z - z_0)$ in (3.5) is known as the *residue* $\text{Res}(f; z_0)$ of f at $z = z_0$.

Definition 3.4 (Poles): An isolated singularity z_0 is called a pole of order m , if the coefficients $a_{-k} = 0$ for $k > m$ and $a_{-m} \neq 0$. A pole of order 1 is a simple pole, poles of order 2 and 3 are double and triple poles respectively.

Theorem 3.3 (Residue at a Pole): An isolated singularity z_0 of a complex function $f(z)$ is a pole of order $m \geq 1$, if and only if $\psi_m(z) = (z - z_0)^m f(z)$ is analytic at z_0 , and

(a) For $m = 1$,

$$\text{Res}(f; z_0) = \lim_{z \rightarrow z_0} \psi_1(z) = \lim_{z \rightarrow z_0} (z - z_0) f(z), \quad (3.6)$$

(b) For $m > 1$,

$$\text{Res}(f; z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \psi_m^{(m-1)}(z). \quad (3.7)$$

Exercise 3.5: Use a Laurent's series expansion to show that the point of singularity of $f(z)$ is a pole. Also, find the order m of the pole and the corresponding residue B :

(a) $\frac{1 - \cos z}{z^3}$

¹ $z_0 \in \mathbb{C}$ is an *isolated singularity* of $f(z)$, provided $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$.

- (b) $\frac{1 - e^{2z}}{z^4}$
 (c) $1/z^2 e^z$
 (d) $\sin z/z^3$
 (e) $\frac{1 - \cosh z}{z^5}$

Answers:

- (a) $\frac{1 - \cos z}{z^3} = \frac{1}{2!} \frac{1}{z} - \frac{z}{4!} + \frac{z^3}{6!} - \frac{z^5}{8!} + \dots$, $\text{Res}(f; 0) = 1/2$
 (b) $\frac{1 - e^{2z}}{z^4} = -\frac{2}{z^3} - \frac{1}{2z^2} - \frac{1}{6z} - \frac{1}{24} - \dots$, $\text{Res}(f; 0) = -1/6$
 (c) $\frac{1}{z^2} e^{-z} = \frac{1}{z^2} \left(1 - \frac{z}{1!} + \frac{z^2}{2!} - \frac{z^3}{3!} + \dots \right)$, $\text{Res}(f; 0) = -1$
 (d) $\frac{\sin z}{z^3} = \frac{1}{z^2} - \frac{1}{3!} + \frac{z^2}{5!} - \frac{z^4}{7!} + \dots$, $\text{Res}(f; 0) = 0$
 (e) $\frac{1 - \cosh z}{z^5} = -\frac{1}{2z^3} - \frac{1}{24z} + \frac{z}{6!} - \frac{z^3}{8!} - \dots$, $\text{Res}(f; 0) = -1/24$

Exercise 3.6: Find the residues of each of the following functions $f(z)$ at its poles:

- (a) $1/(z-1)(z^2+1)$
 (b) $1/(z+z^2)$
 (c) $e^z/(z^2+\pi^2)$
 (d) $e^{2z}/(z-1)^2$
 (e) $1/(z-1)(z+3)^2$
 (f) $\frac{1}{z^2+16}$
 (g) $\left(\frac{z+1}{z-1}\right)^3$
 (h) $1/z(z-1)^2$
 (i) $\left(\frac{z}{2z+1}\right)^3$
 (j) $\frac{z^2+1}{z(z-1)}$

Answers:

- (a) $z = 1, \pm i$ are simple poles of f , $\text{Res}(f; 1) = 1/2$, $\text{Res}(f; i) = \frac{1-i}{4}$, $\text{Res}(f; -i) = \frac{i+1}{4}$
 (b) $z = 0$ and -1 are simple poles of f , $\text{Res}(f; 0) = 1$, $\text{Res}(f; -1) = -1$
 (c) $z = \pm \pi i$ are simple poles of f , $\text{Res}(f; \pi i) = \frac{i}{2\pi}$, $\text{Res}(f; -\pi i) = -\frac{i}{2\pi}$
 (d) $z = 1$ is a double pole of f , $\text{Res}(f; z_1) = 2e^2$

- (e) $z = 1$ is a simple pole and $z = -3$ is a double pole for f , $\text{Res}(f; 1) = 1/16$, $\text{Res}(f; -3) = -1/16$
- (f) $z = \pm 4i$ are simple poles, and $\text{Res}(f; 4i) = -i/8$, $\text{Res}(f; -4i) = i/8$;
- (g) $z = 1$ is a triple pole, and $\text{Res}(f; 1) = 6$;
- (h) $z = 0$ is a simple pole, and $\text{Res}(f; 0) = 1$; $z = 1$ is a double pole, and $\text{Res}(f; 1) = -1$
- (i) $z = -1/2$ is a triple pole, and $\text{Res}(f; -1/2) = 3/16$
- (j) $z = 0, 1$ are simple poles, and $\text{Res}(f; 0) = 2$, $\text{Res}(f; 1) = 2$.

Exercise 3.7: Find the residues of each of the following functions $f(z)$ at its poles:

- (a) $\frac{z^2 + 2}{z - 1}$
- (b) $\frac{1}{z^4 + 1}$
- (c) $\frac{\cos z}{z^2(z - \pi)^3}$
- (d) $\frac{4z + 8}{2z - 1}$
- (e) $1/(z^+z^3 - 2z^2)$.

Answers:

- (a) $z = 1$ is a simple pole and $\text{Res}(f; 1) = 3$
- (b) $z_k = e^{i(2k-1)\pi/4}$ are the simple poles, $\text{Res}(f, z_k) = -z_k/4$ for $k = 1, 2, 3, 4$
- (c) $\text{Res}(f, 0) = -3/\pi^4$, $\text{Res}(f, \pi) = (\pi^2 - 6)/2\pi^4$
- (d) $\text{Res}(f, 1/2) = 5$
- (e) $\text{Res}(f, 0) = -1/4$, $\text{Res}(f, 1) = 1/3$, $\text{Res}(f, -2) = -1/12$.

3.4 Integration around Closed Contours

Theorem 3.4 (Cauchy's Integral Formula): Suppose that f is analytic on a simply connected domain $\mathcal{D}$ and γ , a simple closed contour lying entirely within $\mathcal{D}$. Then for any point a within γ , we have

$$f(a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - a} dz \text{ or } \oint_{\gamma} \frac{f(z)}{z - a} dz = 2\pi i f(a). \quad (3.8)$$

Theorem 3.5 (Cauchy's Integral Formula for Derivatives): Under the assumptions of Theorem 3.4, we have

$$\oint_{\gamma} \frac{f(z)}{(z - a)^{n+1}} dz = \frac{2\pi i}{n!} \cdot f^{(n)}(a) \text{ for } n = 1, 2, 3, \dots \quad (3.9)$$

Exercise 3.8: Compute each of the following integrals around a circle γ , by employing the Cauchy's formulas (3.8) and (3.9):

- (a) $\oint_{\gamma} \frac{\cos z}{z} dz; \gamma : |z| = 1$
- (b) $\oint_{\gamma} \frac{z^2 - z + 1}{z - 1} dz; \gamma : |z| = 3/2$
- (c) $\oint_{\gamma} \frac{\sin z}{(z - \pi/6)^3} dz; \gamma : |z| = 1$
- (d) $\oint_{\gamma} \frac{3z^2 + z}{z^2 - 1} dz; \gamma : |z - 1| = 1$
- (e) $\oint_{\gamma} \frac{z - 1}{(z + 1)^2(z - 2)} dz; \gamma : |z - i| = 3.$

Answers:

- (a) Let $f(z) = \cos z$. Then f is analytic on the whole complex plane $\mathbb{C}$ and $z = 0$ lies within the circle $\gamma : |z| = 1 \Rightarrow \oint_{\gamma} \frac{f(z)}{z} dz = 2\pi i f(0) = 2\pi i$
- (b) Let $f(z) = z^2 - z + 1$. Then f is analytic on $\mathbb{C}$ and $z = 1$ lies within the circle $\gamma : |z| = 3/2 \Rightarrow \oint_{\gamma} \frac{f(z)}{z - 1} dz = 2\pi i f(1) = 2\pi i$
- (c) Let $f(z) = \sin z$. Then f is analytic on $\mathbb{C}$ and $z = \pi/6$ lies within $\gamma : |z| = 1 \Rightarrow \oint_{\gamma} \frac{f(z)}{(z - \pi/6)^3} dz = \frac{2\pi i}{2!} f''(\pi/6) = -\pi i/2$
- (d) The point $z = 1$ lies within the circle $|z - 1| = 1$, whereas $z = -1$ lies outside of it. Therefore, define $f(z) = \frac{3z^2 + z}{z + 1} \Rightarrow \oint_{\gamma} \frac{3z^2 + z}{z^2 - 1} dz = \oint_{|z-1|=1} \frac{f(z)}{z - 1} dz = 2\pi i f(1) = 4\pi i$
- (e) $z = -1$ and $z = 2$ lie inside the circle $|z - i| = 3$, and by partial fractions,

$$\frac{z - 1}{(z + 1)^2(z - 2)} = -\frac{1}{9} \frac{1}{z + 1} + \frac{2}{3} \frac{1}{(z + 1)^2} + \frac{1}{9} \frac{1}{z - 2} \Rightarrow \oint_{|z-i|=3} \frac{z - 1}{(z + 1)^2(z - 2)} dz = 0$$

Exercise 3.9: Compute the following integrals around a circle γ using the formulas (3.8) and (3.9):

- (a) $\oint_{\gamma} \frac{z^2 - 4z + 4}{z + i} dz; \gamma : |z| = 2$
- (b) $\oint_{\gamma} \frac{e^{2z}}{(z + 1)^4} dz; \gamma : |z| = 2$
- (c) $\oint_{\gamma} \frac{z}{z^2 + 9} dz; \gamma : |z - 2i| = 4$
- (d) $\oint_{\gamma} \frac{1}{z^2 + 1} dz; \gamma : |z| = 2$
- (e) $\oint_{\gamma} \frac{e^{-z}}{(z - i1)(z - 2)} dz; \gamma : |z| = 3$
- (f) $\oint_{\gamma} \frac{e^{z^2}}{(z - i)^3} dz; \gamma : |z - i| = 1$

- (g) $\oint_{\gamma} \frac{1}{z^3(z-4)} dz; \gamma : |z| = 2$
- (h) $\oint_{\gamma} \frac{z}{(z+i)^4} dz; \gamma : |z| = 2$
- (i) $\oint_{\gamma} \frac{e^{-z}}{(z+2)^5} dz; \gamma : |z| = 3$
- (j) $\oint_{\gamma} \frac{z^2+4}{z^2-5iz+4} dz; \gamma : |z-3i| = 13/10.$

Answers:

- (a) $2(-4 + 3\pi i)$
- (b) $8\pi i/3e^2$
- (c) 0
- (d) 0
- (e) $2\pi i(e-1)/e^2$
- (f) $-2\pi i/e$
- (g) $-\pi i/32$
- (h) 0
- (i) 0
- (j) -8π

Theorem 3.6 (Cauchy's Residue Theorem): Let $\mathcal{D}$ be a simply connected domain and Γ , a simple (positively oriented) closed contour lying completely within $\mathcal{D}$. If f is analytic on and within inside Γ , except a finite number of singularities $z_1, z_2, \dots, z_n$ within Γ , then

$$\oint_{\Gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f; z_k). \quad (3.10)$$

Exercise 3.10: Using Cauchy's residue theorem, find the integral of $f(z)$, around the given contour γ :

- (a) $\frac{1}{z^3(z+4)}, \gamma : |z+2| = 3$
- (b) $\frac{3z^3}{(z-1)(z^2+9)}, \gamma : |z| = 4$
- (c) $\frac{e^z}{\cos \pi z}, \gamma : |z| = 1$
- (d) $\frac{1}{(z-1)(z+2)^2}, |z| = 5/2$
- (e) $\frac{e^z}{z^4+5z^3}, \gamma : |z| = 2.$

Answers:

- (a) 0
- (b) $6\pi i$
- (c) $4i \sinh \frac{1}{2}$
- (d) 0
- (e) $17\pi i/125$

Exercise 3.11 (Self-check): Using Cauchy's residue theorem, find the contour integral of $f(z)$, around γ :

- (a) $\frac{z}{(z+1)(z^2+1)} dz$; $\gamma : 16x^2 + y^2 = 4$
- (b) $ze^{1/z}$; $\gamma : |z| = 1$
- (c) $\log z$; $\gamma : |z| = 1$
- (d) $\frac{1}{z \sin z}$; $\gamma : |z| = 1$
- (e) $\frac{z^4}{(z+1)(z-i)^2}$; $\gamma : 9x^2 + 4y^2 = 36$.

Exercise 3.12 (Self-check): Using Cauchy's residue theorem, compute the following contour integrals:

- (a) $\oint_{|z|=2\pi} \tan z \, dz$
- (b) $\oint_{|z|=3} \frac{1}{(z-1)^2(z-2)} \, dz$
- (c) $\oint_{|z|=4} \frac{z^3}{(z-1)(z-2)(z-3)} \, dz$
- (d) $\oint_{|z-2|=3/2} \frac{1}{z^3(z-1)^4} \, dz$
- (e) $\oint_{|z|=2} \frac{z}{z^4-1} \, dz$.

3.5 Evaluation of Real Integrals by Cauchy's Residue Theorem

Type 1: $\int_0^{2\pi} Q(\cos \theta, \sin \theta) \, d\theta$

■ Let $z = e^{i\theta}$ be any point on $\gamma : |z| = 1$ so that $dz = ie^{i\theta} d\theta$ or $d\theta = dz/iz$, where $0 \leq \theta \leq 2\pi$.

■ Using Euler's identity: $e^{i\theta} = \cos \theta + i \sin \theta$, we see that $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z^2 + 1}{2z}$ and $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z^2 - 1}{2zi}$. Therefore,

$$\int_0^{2\pi} Q(\cos \theta, \sin \theta) \, d\theta = \oint_{\gamma} \underbrace{\frac{1}{iz} \cdot Q\left(\frac{z^2 + 1}{2z}, \frac{z^2 - 1}{2zi}\right)}_{=f(z)} \, dz. \quad (3.11)$$

- Find the residues of $f(z)$ at its poles $z_1, z_2, \dots, z_n$ lying within γ , and then employ Cauchy's Residue Theorem.

Exercise 3.13: Evaluate the following real integrals:

$$(a) \int_0^{2\pi} \frac{\sin \theta}{5+4 \cos \theta} d\theta$$

$$(b) \int_0^{2\pi} \frac{1}{2+\sin \theta} d\theta$$

$$(c) \int_0^{2\pi} \frac{d\theta}{1+3 \cos^2 \theta}$$

$$(d) \int_0^{2\pi} \frac{d\theta}{a+\cos \theta}, \text{ where } a > 1$$

$$(e) \int_0^{2\pi} \frac{d\theta}{(5-3 \cos \theta)^2}$$

$$(f) \int_0^{2\pi} \frac{1+2 \cos \theta}{5+4 \cos \theta} d\theta$$

$$(g) \int_0^{2\pi} \frac{1}{2-\cos \theta} d\theta$$

$$(h) \int_0^{2\pi} \frac{1}{1+\sin^2 \theta} d\theta$$

$$(i) \int_0^{2\pi} \frac{\cos \theta d\theta}{3+\sin \theta}$$

$$(j) \int_0^{2\pi} \frac{1}{3+\cos^2 \theta} d\theta$$

Answers:

$$(a) 0$$

$$(b) 2\sqrt{3} \pi/3$$

$$(c) \pi$$

$$(d) \frac{2\pi}{\sqrt{a^2-1}}$$

$$(e) 5\pi/2$$

$$(f) 0$$

$$(g) 2\pi\sqrt{3}/3$$

$$(h) \pi\sqrt{2}/2$$

$$(i) 0$$

$$(j) \pi$$

Type 2: $\int_{-\infty}^{\infty} f(x) dx$: Let $f(x)$ be a rational function such that its complex form $f(z)$ has a finite number of poles above the real axis.

Recall that $I = \int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx$, provided that the limit on the right hand side exists. To evaluate I , we choose a closed contour $\gamma = [-R, R] \cup \gamma_R$, where

■ $[-R, R]$ is the interval of length $2R$ on the real axis, $R \gg 0$ and

■ γ_R is the arc of the semicircle $|z| = R$ from $(R, 0)$ to $(-R, 0)$, in the anticlockwise direction in the upper half plane. Along γ_R , $z = e^{it}$, $0 \leq t \leq \pi$.

Then

$$\oint_{\gamma} f(z) dz = \int_{-R}^R f(x) dx + \int_{\gamma_R} f(z) dz. \quad (3.12)$$

We find the singularities $z_1, z_2, \dots, z_n$ of f , which lie inside γ (of course above the real line - why?), and hence the residues there at. Then by residue theorem,

$$\oint_{\gamma} f(z) dz = 2\pi i \cdot \sum_{j=1}^n \text{Res}(f; z_j). \quad (3.13)$$

Employing the limit as $R \rightarrow \infty$ in (3.12), and using (3.13), we find that $I = 2\pi i \cdot \sum_{j=1}^n \text{Res}(f; z_j)$, provided $\left| \int_{\gamma_R} f(z) dz \right| \rightarrow 0$ as $R \rightarrow \infty$.

Exercise 3.14: Evaluate the following real integrals:

- (a) $\int_0^{\infty} \frac{dx}{x^2+1}$
- (b) $\int_0^{\infty} \frac{dx}{(x^2+a^2)^2}, a > 0$
- (c) $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)} dx$
- (d) $\int_{-\infty}^{\infty} \frac{1}{x^2-2x+2} dx$
- (e) $\int_{-\infty}^{\infty} \frac{x^2-x+2}{x^4+10x^2+9} dx$
- (f) $\int_0^{\infty} \frac{x^2+1}{x^4+1} dx$
- (g) $\int_{-\infty}^{\infty} \frac{1}{(x^2+1)^3} dx$
- (h) $\int_0^{\infty} \frac{1}{x^6+1} dx$
- (i) $\int_{-\infty}^{\infty} \frac{1}{x^2-2x+25} dx$
- (j) $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2} dx$

Answers:

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4a^3}$

(c) $\frac{\pi}{2\sqrt{2}}$

(d) π

(e) $\frac{5\pi}{12}$

(f) $\frac{\pi\sqrt{2}}{2}$

(g) $3\pi/8$

(h) $\frac{\pi}{6}$

(i) $\pi\sqrt{6}/12$

(j) $\pi/2$

Type 3: $\int_{-\infty}^{\infty} Q(x) \cos ax \, dx$ or $\int_{-\infty}^{\infty} Q(x) \sin ax \, dx$, where $Q(x)$ is a rational function having no poles on the real axis.

Exercise 3.15 (Type III: Poles Lying on the Real Axis): Evaluate

(a) $\int_0^{\infty} \frac{\cos 3x}{x^2+4x+5} \, dx$

(b) $\int_{-\infty}^{\infty} \frac{\sin x}{x^2+4x+5} \, dx$

(c) $\int_0^{\infty} \frac{x \sin mx \, dx}{x^2+a^2}, a > 0, m > 0$

(d) $\int_0^{\infty} \frac{x \sin x}{(x^2+1)(x^2+4)} \, dx$

(e) $\int_{-\infty}^{\infty} \frac{\cos x}{(x^2+1)(x^2+9)} \, dx$

Answers:

(a) $\frac{2\pi}{e^3}$

(b) $-\frac{\pi \sin 2}{e}$

(c) $\frac{\pi}{2}e^{-am}$

(d) $\frac{\pi}{6}(e^{-1} - e^{-2})$

(e) $\frac{1}{8e} - \frac{1}{24e^3}$

Type 4: Indented Contours

Theorem 3.7: Suppose that $f(z)$ has a simple pole $z = c$ on the real axis. If γ_ϵ is the contour in the parametric form $z = c + \epsilon e^{i\theta}$ for $0 \leq \theta \leq \pi$, then

$$\lim_{\epsilon \rightarrow 0} \int_{\gamma_\epsilon} f(z) dz = \pi \cdot \text{Res}(f; c) \quad (3.14)$$

Exercise 3.16: Evaluate the following integrals through indented contours:

(a) $\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2-2x+2)} dx$

(b) $\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx$

(c) $\int_0^{\infty} \frac{1-\cos x}{x^2} dx$

(d) $\int_0^{\infty} \frac{\sin x}{x} dx = \pi/2$

(e) $\int_0^{\infty} \frac{\sin 3x}{x} dx$

Answers:

(a) $\frac{\pi}{2} \left(1 + \frac{\sin 1 - \cos 1}{e} \right)$

(b) $\pi \left(1 - \frac{1}{e} \right)$

(c) $\pi/2$

(d) $\pi/2$

(e) $\pi/2$

Glossary of Terms

Binomial Expansion Formula: $(1 + \eta)^\alpha = 1 + \alpha\eta + \frac{\alpha(\alpha-1)\eta^2}{2!} + \frac{\alpha(\alpha-1)(\alpha-2)\eta^3}{3!} + \dots$ for $|\eta| < 1$, $\alpha \in \mathbb{R}$

Taylor's Series about a Point: Let $f(z)$ be analytic on $\mathcal{D} \subset \mathbb{C}$ and $z_0 \in \mathcal{D}$. Then the Taylor's series of f about z_0 is given by $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots$, where $a_0 = f(z_0)$, $a_n = \frac{f^{(n)}(z_0)}{n!}$, $n \geq 1$.

Isolated Singularity: If $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$, then z_0 is called an *isolated singularity* of f . If $f(z) = \phi(z)/\psi(z)$, where $\psi(z_0) = 0$ and $\phi(z_0) \neq 0$, then z_0 is an isolated singularity of f

Laurent's series about an Isolated Singularity: Let z_0 be an isolated singularity of f . Suppose that f be analytic on the annulus $\mathcal{A} : 0 < r < |z - z_0| < R$. The Laurent's series of f about z_0 is given by

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n} + \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

Residue at an Isolated Singularity: The coefficient a_{-1} of the first negative power $1/(z - z_0)$ in the Laurent's series expansion of f is known as the *residue* $\text{Res}(f; z_0)$ of f at $z = z_0$. The residue should be computed at an isolated singularity, which lies within the region of validity of the Laurent's series. The residue is not defined at an isolated singularity not lying in the validity region.

Poles: An isolated singularity z_0 is called a pole of order m , if the coefficients $a_{-k} = 0$ for $k > m$ and $a_{-m} \neq 0$. A pole of order 1 is a simple pole, poles of order 2 and 3 are double and triple poles respectively.

Residue at a Simple Pole: If z_0 is a simple pole of $f(z)$, then $\text{Res}(f; z_0) = \lim_{z \rightarrow z_0} \psi_1(z) = \lim_{z \rightarrow z_0} (z - z_0)f(z)$.

Residue at a Multiple Pole: If z_0 is a pole of order $m > 1$ of $f(z)$, then $\text{Res}(f; z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \psi_m^{(m-1)}(z_0)$.

Cauchy's Integral Formula: Suppose that f is analytic on a simply connected domain $\mathcal{D}$ and γ , a simple closed contour lying entirely within $\mathcal{D}$. Then for any point a within γ , $\oint_{\gamma} \frac{f(z)}{z-a} dz = 2\pi i f(a)$.

Cauchy's Integral Formula for Derivatives: Suppose that f is analytic on a simply connected domain $\mathcal{D}$ and γ , a simple closed contour lying entirely within $\mathcal{D}$. Then $\oint_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} \cdot f^{(n)}(a)$ for $n = 1, 2, 3, \dots$ for any point a within γ .

Cauchy's Residue Theorem: Let $\mathcal{D}$ be a simply connected domain and Γ , a simple (positively oriented) closed contour lying completely within $\mathcal{D}$. If f is analytic on and within inside Γ , except a finite number of singularities $z_1, z_2, \dots, z_n$ within Γ , then $\oint_{\Gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f; z_k)$.

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